

STEPANOVKA, Russian Federation

WMO: 291490

Lat: 58.640N

Lon: 86.730E

Elev: 344

StdP: 14.51

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.2	(c) -36.3	(d) -47.1	(e) 0.4	(f) -41.4	(g) -40.9	(h) 0.5	(i) -35.2	(j) 18.4	(k) 8.2	(l) 16.1	(m) 13.7	(n) 2.1	(o) 20	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	84.9	65.2	81.7	63.3	78.6	62.7	68.3	79.4	66.4	76.7	64.7	74.5	5.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	92.7	72.1	62.8	86.6	70.3	61.0	81.0	69.2	32.9	79.2	31.4	77.0	30.1	74.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.1	(b) 14.2	(c) 12.1	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 17.1	(b) 14.2	(c) 12.1	(d) WB	(e) -45.9	(f) 71.3	(g) 3.8	(h) 1.7	(i) -48.7	(j) 72.5	(k) -50.9	(l) 73.5	(m) -53.1	(n) 74.4	(o) -55.9	(p) 75.6	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.1	-4.4	1.6	18.6	33.5	45.3	60.3	65.3	59.4	47.3	31.8	10.4	1.6	
	DBStd	26.81	16.52	15.70	12.48	9.77	9.28	9.64	6.07	6.42	8.21	9.68	16.10	17.73	
	HDD50	8155	1687	1355	973	498	210	27	1	4	147	566	1187	1500	
	HDD65	12553	2152	1775	1438	946	612	199	71	195	533	1030	1637	1965	
	CDD50	1239	0	0	0	2	65	334	475	296	65	2	0	0	
	CDD65	164	0	0	0	0	3	57	81	22	1	0	0	0	
	CDH74	1896	0	0	0	1	61	717	878	222	17	0	0	0	
	CDH80	465	0	0	0	0	7	207	219	31	1	0	0	0	
Wind	WSAvg	5.2	4.7	4.9	5.8	6.3	6.3	5.3	4.4	4.2	4.5	5.4	5.8	5.0	
Precipitation	PrecAvg	26.7	1.6	1.3	1.5	1.7	2.2	2.6	2.8	2.8	2.4	2.8	2.8	2.2	
	PrecMax	34.9	3.3	2.3	3.3	3.0	5.0	4.2	4.8	4.9	3.6	4.1	4.4	4.0	
	PrecMin	18.9	0.1	0.4	0.3	0.6	0.9	0.6	0.5	1.2	0.7	1.1	1.5	0.7	
	PrecStd	3.7	0.7	0.5	0.7	0.6	0.9	0.8	1.1	1.0	0.7	0.7	0.8	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.6	33.8	46.3	61.7	80.7	90.3	88.2	83.9	75.8	60.0	40.9	32.4	
		MCWB	29.4	30.9	36.2	46.2	58.1	64.9	69.2	66.7	60.9	47.7	38.0	31.2	
	2%	DB	26.7	30.5	41.8	56.9	74.0	85.2	85.4	79.4	69.6	53.3	34.4	30.1	
		MCWB	25.4	28.7	34.0	44.7	55.2	63.2	67.1	64.6	58.1	45.2	32.7	29.3	
	5%	DB	22.8	27.2	38.3	52.2	67.5	81.6	82.3	76.1	64.4	49.1	32.2	27.4	
		MCWB	21.8	25.6	32.0	41.4	52.4	62.1	64.3	63.2	55.7	43.8	31.1	26.4	
	10%	DB	19.0	23.1	35.1	47.9	61.7	77.9	79.2	72.4	59.6	45.2	29.7	23.8	
		MCWB	18.1	21.3	31.0	38.9	50.0	61.4	63.4	60.9	52.8	40.9	28.5	22.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.4	31.5	38.5	49.2	60.1	68.4	72.6	68.5	62.1	50.0	38.3	31.6
			MCDB	30.1	33.4	45.0	58.5	75.6	83.2	81.6	79.0	72.5	54.4	41.4	32.1
2%		WB	25.5	28.7	35.0	45.7	57.3	66.2	69.6	66.4	59.2	46.7	33.1	29.2	
		MCDB	26.5	30.5	39.4	54.5	69.8	79.0	80.5	75.5	68.5	53.2	34.1	30.0	
5%		WB	21.7	25.7	33.0	42.8	54.3	64.4	67.6	64.8	56.3	43.6	31.1	26.5	
		MCDB	22.7	27.2	37.1	50.9	66.2	77.7	77.5	72.8	63.3	48.6	32.0	27.4	
10%		WB	18.0	21.4	31.1	39.9	50.9	62.4	65.9	62.9	53.6	40.9	28.5	22.9	
		MCDB	19.0	23.0	35.1	46.3	59.8	74.4	75.5	69.9	58.6	45.2	29.7	23.9	
Mean Daily Temperature Range	5% DB	MDBR	15.8	17.5	19.7	17.7	17.6	19.9	20.3	19.1	15.5	10.9	12.9	13.9	
		MCDBR	17.3	11.4	20.1	22.8	26.2	26.9	25.9	25.0	22.3	17.1	10.8	13.0	
		MCWBR	16.6	10.1	14.3	13.5	12.6	9.0	9.9	11.6	12.1	11.2	10.1	12.8	
	5% WB	MCDBR	17.4	11.3	15.3	20.4	23.3	22.5	21.7	21.2	20.8	16.3	11.2	12.8	
		MCWBR	16.8	10.2	11.5	13.6	12.5	8.6	10.5	11.0	12.1	11.4	10.8	12.7	
Clear-Sky Solar Irradiance	taub	0.279	0.315	0.356	0.378	0.396	0.399	0.444	0.407	0.365	0.336	0.295	0.276		
	taud	1.976	1.980	1.949	2.075	2.257	2.340	2.192	2.286	2.356	2.261	2.035	1.925		
	Ebn at Noon	156	211	239	259	264	266	248	250	242	209	148	111		
	Edh at Noon	18	30	41	43	39	36	41	35	28	23	17	14		
All-Sky Solar Radiation	RadAvg	185	488	947	1406	1540	1808	1665	1268	764	375	175	101		
	RadStd	21	40	62	109	158	182	138	98	106	43	37	18		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page